Pavankumar P S

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Linkedin | GitHub | LeetCode

Objective

Dedicated Software Development Engineer with strong proficiency in **Java, Python, and SQL**, focused on building efficient and scalable software solutions. Experienced in **HTML, CSS, and JavaScript** for creating responsive and user-friendly web interfaces. Familiar with **machine learning** and **deep learning** concepts, including **data analysis, visualization**. Passionate about applying data-driven approaches to solve real-world problems and continuously learning to develop innovative solutions.

Education

Dayananda Sagar University — Bengaluru, India
Computer Science and Engineering — December 2022 - Present
CGPA: 9.32

Skills

- Programming Languages: Java, Python and C.
- Web Development: HTML, CSS, JavaScript.
- Database Management: MySQL.
- Machine Learning & Data analysis.
- Tools/Platforms: Tableau, GitHub, VS Code, Google Colab, Linux, Excel.

Coursework

Data Structure, Algorithm & complexity analysis, DBMS, Operating System, OOP, Computer network

Projects

Driver Drowsiness Detection & Alert System — Python, OpenCV, dlib, Twilio API, Google Maps API

- Developed a real-time Driver Drowsiness Detection system using OpenCV and face landmark-based EAR (Eye Aspect Ratio) analysis to monitor alertness levels through webcam video feed.
- Implemented continuous frame-wise detection with dlib and face_recognition libraries, calculating EAR thresholds to trigger alerts when drowsiness is detected.
- Integrated Twilio API for SMS alerts and added Pygame audio warnings, enhancing system responsiveness and driver safety.

Social Media Comment Analyser — Python, Streamlit, Pandas, WordCloud, Matplotlib.

- Built an AI-powered web application using Streamlit to analyze YouTube video comments for sentiment**,** toxicity**,** and frequent word patterns.
- Implemented sentiment classification using a pretrained transformer model**,** and toxicity detection using a multi-label Logistic Regression model with TF-IDF vectorization.
- Enabled users to paste a YouTube video link, view results through bar charts and word clouds, and download analysed comments as a CSV**.**

Money Tracker Web — HTML, CSS, JavaScript, Node.js, MongoDB

- Developed a web-based personal finance application to help users effectively track and manage income and expenses.
- Designed an intuitive interface for managing transactions and implemented real-time financial summaries with transaction history views to enhance usability, engagement, and promote better budgeting and spending habits.

Achievements

- Completed project-based internship tasks at Bharat Intern and received a certificate for developing a weather app and portfolio website.
- Co-authored and presented the research paper "Smart Cradle: A Secured Assistance and Monitoring System for Baby Using IoT" at ICICNIS 2024 International Conference, Bangalore.
- Career Essentials in Generative Al Certification **(Microsoft)**
- The Complete 2024 Web Development Boot camp **(Udemy)**